

Ideas for bounding sample sizes for Belgian RLAs

October 1, 2026

1. Introduction

The Belgian federal elections select 150 members for the Chamber of Representatives ($\approx$ lower house) of the Federal Parliament, following a process that is described in the "Code électoral".¹ These 150 Representatives come from 11 electoral districts, that run 11 independent elections, each for a fixed number of representatives that currently ranges from 4 to 24 as a function of the population of the district (these numbers are adjusted every ten years, following a process defined at Art. 63 of the Belgian constitution).

Each district proposes a ballot to its voters, on which candidates are grouped into parties. In order to vote, a voter can either pick a party or approve as many candidates as they want within a single party (Art. 144 of the "Code électoral"). An example of ballot is visible in Fig. 1. Approving candidates across multiple parties makes an invalid ballot.

The tallying operations are performed independently in each electoral district. Each counting office reports the number of votes that each party received, which are added at the district level. The only parties that can receive a seat are those that obtained the minimum electoral threshold of 5% of the total number of valid votes (Art. 165bis), that is, ballots on which a preference is expressed for one single party (either for the party or by approving candidates on the party) — blank votes are not considered valid (Art. 157).

The seats that are available for the district are then attributed according to the D'Hondt method: the totals of each party that is above the electoral threshold are divided by 1, 2, 3, etc. up to the number of candidates in the party, resulting in lists of quotients, which are merged in descending order. If s seats are to be attributed in a district, then each party receives a number of seats that equals the number of appearance of its quotients within the top s quotients of the merged list (Art. 167). We illustrate this process in Fig. 2.

In case of equality of quotients, the seat is attributed to the party that has the highest total number of votes. If that is not sufficient to solve the tie, then the candidates who

¹<https://elections.fgov.be/legislation/lois>, last updated on Oct. 27, 2023.

would get the seats are compared: the seat goes to the party of the candidate who has the highest number of preferential votes or, if it is not enough to break the tie, to the party of the oldest candidate (Art. 168). We do not discuss the fairly complex process that determines how the seats are attributed to the candidates: it is irrelevant to the present paper.

2. Auditing the Belgian Federal Elections

2.1. Context of the Audit

Approximately half of the Belgian voters vote on hand-marked and hand-counted paper ballots. Voting happens on a Sunday morning, and counting offices operate on Sunday afternoon and sometimes evening. Each counting office has 6 members, who may tally up to 2400 ballots. The ballots are sorted by parties, then counted, then placed into sealed envelopes. All the ballots in any single envelope must express votes for the same party.

The other half of the voters vote on ballot marking devices that print paper ballots, which are scanned, and counted from the scanned electronic records. Our central focus here is to evaluate the feasibility of running a risk limiting audit for such elections. The audit would be based on an evolution of the current BMDs that has been proposed by a consortium of universities at the request of the Belgian federal ministry of interior, which is in charge of selecting voting systems that can be used in Belgium. The goal of the audit is to limit the risk that seats are granted to the wrong parties. As we will see, the election margins tend to be very tight in Belgium, and auditing the seat allocation to the candidate is expected to make audits impractical – this deserves further explorations.

2.2. The election manifest

The RLA starts from an election manifest that indicates, among other things, how many ballots have been cast, how many votes each party received, and how many votes are invalid – this last category includes all the votes that do not express support for one and only one party and includes the blank votes.

The manifest typically contains other book keeping information, including a list of boxes and envelopes and the reported content of each envelope. In the context of hand-counted ballots, each envelope contains ballots supporting a single party, and invalid ballots are also gathered in one envelope. In the context of e-voting, each envelope contains ballots of which an interpretation has been reported during the scanning process. The important observation here is that both the hand marked and the machine marked ballots are stored in a way that offers a commitment on the content of every ballot in every envelope, which can support efficient ballot-level comparison audits. [Yes, though it might be a bit fiddly to find the n-th card in the envelope. Are they marked in any way?](#)

VT

2.3. Auditing D'Hondt Elections

Let L be a party list. We introduce the following notations and definitions.

- S is the number of available seats.
- $d(i)$ is the divisor for the i -th index. For D'Hondt, $d(i) = i$.
- $\mathcal{P}$ is the set of parties.
- If $P \in \mathcal{P}$ is a party, $\|P\|$ is the number of candidates fielded by P .
- $s :: \mathcal{P} \rightarrow \mathbb{N} \cup \{0\}$ is an allocation of seats to parties.
 $s(P)$ is both the number of seats allocated to party P and (if non-zero) the index of the candidate in party P who is reported to get seated with the lowest quotient. This must satisfy $s(P) \leq \|P\|$.
- The *tally* $\mathcal{T}(P)$ is the total number of valid votes for P .
- The *quotient* of the i -th ranked candidate of a party P , denoted $\mathcal{Q}(L, i)$, is the list's tally divided by $d(i)$.

In a standard D'Hondt election, when there are s seats, Party P gets at least i seats if $\mathcal{Q}(P, i)$ is within the highest s quotients across all parties and if party P has at least i candidates. We still need a precise statement of the function, for example if some party lists exhaust but not others, what exactly is the correctness condition? VT

2.3.1. D'Hondt assertions

Let A and B be two party lists in the election.

Notation Define the generalised D'Hondt assertion as

$$DH_{A,B}(w, l) \iff \mathcal{Q}(A, w) > \mathcal{Q}(B, l).$$

That is,

$$DH_{A,B}(w, l) \iff \frac{\mathcal{T}(A)}{d(w)} > \frac{\mathcal{T}(B)}{d(l)}.$$

Usually we check only that the last winner of party A won over the first loser of party B (if B had any losers). This is vacuously true if all B 's candidates got seats. The following shorthand captures this case.

$$DH_{A,B} = \begin{cases} \text{FALSE}, & \text{if } s(A) = 0, \\ DH_{A,B}(s(A), s(B) + 1), & \text{if } s(A) > 0 \text{ and } s(B) < \|B\| \\ \text{TRUE}, & \text{otherwise.} \end{cases}$$

Note that this assertion is not symmetric (nor antisymmetric): $DH_{A,B}$ does not imply $DH_{B,A}$ (nor $\neg DH_{B,A}$) and, depending on the specifics of an election, these two assertions

may both need to be checked. For example, we may need to check both “Party A ’s 2nd candidate beats Party B ’s 4th candidate,” and also “Party B ’s 3rd candidate beats Party A ’s 3rd candidate” which would be part of proving that A got 2 seats and B got 3.

From now on we will ignore the cases in which $s(A) = 0$ or $s(B) = \|B\|$, in which $DH_{A,B}$ does not need to be audited,² and consider how to audit this assertion when $s(A) > 0$ and $s(B) < \|B\|$.

Following Blom et al. [1, Section 5.2], we can define the proto-assorter of a ballot b for this assertion as

$$g_{A,B}(b) = \frac{b_A}{s(A)} - \frac{b_B}{s(B) + 1},$$

where b_P is set to 1 if the ballot b supports party P , and b_P is set to 0 otherwise.

We can turn this proto-assorter into an assorter

$$h_{A,B}(b) = b_A \cdot \frac{s(B) + 1}{2s(A)} + \frac{1 - b_B}{2}.$$

We can verify that $h_{A,B}(b)$ is 0 for a vote for B , is $1/2$ for a vote for neither A nor B , and is $\frac{s(B)+1}{2s(A)} + \frac{1}{2}$ for a vote for A . Furthermore, when the tally is computed on a set of ballots $\mathcal{B}$, we will have

$$DH_{A,B} \iff \sum_{b \in \mathcal{B}} h_{A,B}(b) / |\mathcal{B}| > 1/2.$$

For an ordinary D’Hondt election with a set $\mathcal{B}$ of ballots, the reported outcome would be correct if the expected number of seats have been granted and if $DH_{A,B}$ is true for every ordered pair of distinct parties A and B .

2.4. Adding a threshold

In Belgian federal elections, parties can obtain seats only if they obtained at least 5% of the valid votes. We use the following notations for capturing this threshold mechanism. $\Sigma_{P \in \mathcal{UP}} \mathcal{T}(P)$ is the *number of valid votes*.

Definition 1 (Threshold). *The electoral threshold is*

$$\theta = \Sigma_{P \in \mathcal{UP}} \mathcal{T}(P) \times 5/100.$$

TODO: θ is defined here as an absolute count but used below as a fraction. Fix Def.

We write the assertion that the party P is above the threshold as

$$AT(P) \iff \mathcal{T}(P) \geq \theta.$$

The threshold only has to be equalled, but our assertions are by convention strict. VT
Comment. Similarly, $BT(P)$ is the assertion that the party P is below the threshold,

²We may need to audit other assertions in order to confirm that $s(A) = 0$ or $s(B) = \|B\|$.

that is $BT(P) = \neg AT(P)$. Again, we use these notations both as names for the assertions and for the logical statement that the assertion is true.

We can turn this into an assorter using the supermajority formulation from SHANGRLA.

$$above\text{-}\theta_A(b) = \frac{b_A}{2\theta} + \frac{b_{INV}}{2},$$

where b_{INV} is 1 if b is invalid, zero otherwise. VT
 It's there so that invalid votes are neutral rwt threshold-testing. TODO check exactly what the threshold-passing rule is - is it 5% of all votes or 5% of valid votes?

We can verify that $above\text{-}\theta_A(b)$ is 0 for a vote for some other party, is $1/2$ for invalid votes, and is $\frac{b_A}{2 \times \theta}$ for a vote for A ($= 10$ in our case). Furthermore, when the tally is computed on a set of ballots $\mathcal{B}$, we will have

$$AT(A) \iff \sum_{b \in \mathcal{B}} above\text{-}\theta_A(b) / |\mathcal{B}| > 1/2.$$

There are far too many as,bs,As, and Bs floating around here. In the below, A is a party, b is a ballot, and bt is the name of an assorter. Similarly, the assorter for $BT(A)$ is: VT

$$below\text{-}\theta_A(b) = \frac{1 - b_A}{2(1 - \theta)} + \frac{b_{INV}}{2}$$

Devin says this is slightly wrong: “ $bt_A(b)$ (line 355): for an invalid ballot $b_A = 0$, so VT
 the first term is $1/(2(1 - \theta))$, plus $b_{INV}/2$ — about 1.03, not the neutral $1/2$. Needs $(1 - b_A - b_{INV})/(2(1 - \theta)) + b_{INV}/2$. (at_A is fine.) ” This is the opposite of at : it is 0 when b is a vote for A . If the valid votes comprise x votes for A and $x(1 - \theta)$ votes for other parties, then the total is exactly $1/2$. So we have

$$BT(A) \iff \sum_{b \in \mathcal{B}} below\text{-}\theta_A(b) / |\mathcal{B}| > 1/2.$$

Formula from JC, with an extra term for invalid votes. VT

2.5. Summary: checking the election outcome

Proposition 1. *In a Belgian federal election, we define $\mathcal{W}$ the set of parties that have at least one reported winner and $\mathcal{L}$ the set of parties that have at least one reported loser, that is $s(L) < \|L\|$. (Some parties may be in both $\mathcal{W}$ and $\mathcal{L}$.) The reported outcome is correct if we prove:*

$$AT(A) \quad \text{for all } A \in \mathcal{W} \quad (1)$$

$$BT(B) \bigvee \left(\bigwedge_{A \in \mathcal{W}} DH_{A,B} \right) \quad \text{for all } B \in \mathcal{L}, B \neq A. \quad (2)$$

There are also basic validity conditions: $\sum_{P \in \mathcal{P}} s(P) = S$ and $\forall P \in \mathcal{P}, s(P) \leq |P|$. These are not assertions that need to be audited, but validity conditions to be checked during the audit.

Proof. This matches the `DHondt.complete` data structure in LEAN. We have `complete_assertion_claim_implies` and some theorems proving the equivalence of this characterisation with the procedural allocation algorithm. $\square$

Figure 3 shows an example of the assertions described in Proposition 1.

2.5.1. Assigning all seats

If the reported outcome does not assign all the seats, then the result is correct iff, for every party P ,

$$s(P) = \begin{cases} |P| & \text{if } AT(P), \\ 0, & \text{otherwise.} \end{cases}$$

It would help to explain this case in the intro—I’m not quite sure what the rules are VT
here. In other words, non-assigned seats can only happen when the losing candidates are coming from parties that did not meet the threshold. Furthermore, every list meeting the threshold must have all of its candidates winning.

In this case Proposition 1 reduces to:

1. For all $P \in \mathcal{L}$: $BT(P)$
2. For all $P \notin \mathcal{L}$: $AT(P)$

No, I think (if I understand the rules correctly, which quite possibly I don’t) that if there are more seats than candidates whose parties get over the threshold, then everyone over the threshold gets all their candidates seated, and that’s it.

AE don’t we still need to prove DH assertions in this case?

VT

3. Which assertions should be checked?

When some seats are unassigned, the verification process is clear: we need to check all the AT and BT assertions, one per party. This case is simple and does not happen in practice.

When all the seats are assigned, there are $2^{|\mathcal{L}|}$ “useful” sets of assertions to check: for each $B \in \mathcal{L}$, we can choose to verify either $BT(B)$ or $\bigwedge_{A \in \mathcal{W}} DH_{A,B}$.

The best way to choose is probably to estimate the sample size and choose the option with the lower estimate. Appendix A contains some possible heuristics, and some discussion of using alternative processes more like AWAIRE. VT

From John C: Using the smallest margin won’t give you the highest difficulty solving, one needs to use `noerror`, which takes into account the upper limit of the assorter, which is not 1 for all the assorters (DH , AT , BT). `noerror = 1.0/(2.0-assorterMargin/assorter.upperBound())` see details at <https://github.com/JohnLCaron/rlauxe/blob/main/docs/Dhondt.md#31-check-margins>

For now, we simply assume that we have some way of choosing which of $BT(B)$ or $\bigwedge_{A \in \mathcal{W}} DH_{A,B}$ is likely to have a smaller sample size.

	LES					BELG.				COLLECTIF	LUTTE	CHEZ
	MR	PS	ENGAGÉS	PTB	N-VA	DéFI	ECOLO	UNIE-BUB	CITOYEN	OUVRIERE	NOUS	BLANCO
Seats	5	6	3	3	0	0	0	0	0	0	0	0
%	26	29	15	14	1.9	2.1	5	0.33	0.97	0.63	3.0	1.8
Votes	192759	213501	114559	103339	14184	15189	36750	2426	7201	4680	22039	13224
/2	96380	106751	57280	51670	7092	7595	18375	1213	3601	2340	11020	6612
/3	64253	71167	38186	34446	4728	5063	12250	809	2400	1560	7346	4408
/4	48190	53375	28640	25835	3546	3797	9188	607	1800.25	1170	5510	3306
/5	38552	42700	22912	20668	2837	3038	7350	485	1440.2	936	4408	2645
/6	32127	35584	19093	17223	2364	2531.5	6125	404	1200	780	3673	2204
/7	27537	30500	16366	14763	2026	2170	5250	347	1029	669	3148	1889

Table 1: Election outcome for Hainaut, 2024, in which 17 seats were allocated. Seated quotients are shown in blue. The threshold, at 5%, is 36993, which the Ecolo party very narrowly falls short of.

4. Relaxing verification for hard cases

Sometimes a single assertion is so hard to prove (because the margin is so close) that auditing is infeasible. In these cases, we may not be able to verify the exact assignment of all seats to all parties. In this section we devise assertions for proving part of an election outcome, for use when proving the entire election outcome correct is infeasible. Because the number of seats is fixed, an uncertainty about one seat usually implies (the opposite) uncertainty about at least one other.

VT

TODO: I'm leaving out the case in which not all seats are assigned, because I think it's straightforward, being based entirely on thresholds: just audit the thresholds you can, and admit that you can't audit the others.

4.1. Motivating example

Consider the outcome of the circonscription of Hainaut, 2024. Apparently, Ecolo fails the threshold. However, the margin by which this occurs is so tiny as to be infeasible to audit. The results are shown in Table 1, with assigned seats in blue.

We'll have to fix this up when we correctly calculate the threshold as a function of the total, not the valid ones - TODO. Actually our intro says it's a function of the valid ones. TODO check.

VT

Auditing the whole election outcome is infeasible because $BT(\text{Ecolo})$ is infeasible. Ecolo's first candidate is not a D'Hondt loser, so that alternative audit option does not work. If we simply omit the assertion $BT(\text{Ecolo})$ from the equations in Proposition 1, then we leave open the possibility that Ecolo might have won far more than 1 seat. If it is true that actually $AT(\text{Ecolo})$, then (apparently) PTB would miss out on its 3rd seat, which is the last allocated. We cannot therefore hope to verify that PTB got 3 seats without verifying $BT(\text{Ecolo})$. The best we can hope for is to verify

- that PTB got 2 or 3 seats,
- that Ecolo got 0 or 1 seat,
- that all other parties got exactly their apparent allocation.

That is, either Ecolo passed the threshold and got a seat, and PTB got 2 seats, or Ecolo failed the threshold and got no seats, and PTB got 3 seats.

In this example, Ecolo would get only one seat if it crossed the threshold, though this is not true in general. It is always true that the last seat allocated (i.e. the smallest number) is the one that is lost. [It may not always be the case that the borderline party's second candidate is a D'Hondt loser when it passes the threshold, but we can deal with that case later.](#) VT

4.2. Equivocating over whether a party failed the threshold

Suppose we have a party R that is too close to the threshold to verify. We wish to relax the conditions in Proposition 1 so that everything else is verified.

When a party R has narrowly failed the threshold, and we wish to relax the assertions to allow that it might have passed (or vice versa), then we still need to prove that all the (other) apparently-winning parties passed the threshold and all the apparent losers (other than R) lost. We also need to think carefully about which DH assertions are required.

Define the following partition of $\mathcal{P}$:

AT are those (clearly, verifiably) *above* the threshold—we include an AT assertion for each party in this set;

ET are those too close to the threshold to verify—we include neither an AT nor a BT assertion for parties in this set;

BT are those (clearly, verifiably) *below* the threshold—we include a BT assertion for each party in this set.

The discussion of which ones we decide are “too close” is a separate question. Again we simply assume that we have some way of estimating sample sizes, and some judgement about which ones are acceptable. In this section, we simply specify how we state the relaxation, and which assertions will imply a given relaxation.

Useful logical fact If X, Y, Z are logical propositions, the following is true:

$$Y \wedge Z \implies (X \wedge Y) \vee (\neg X \wedge Z)$$

We will set X to be the assertion $AT(R)$ and let Y and Z respectively be all the other things we would need to check when $AT(R)$ is believed true (resp false). In this way, we prove that either ($AT(L)$ true and everything else is correct) or ($AT(R)$ is false and everything else is correct). This same idea generalises to multiple simultaneous uncertain propositions:

If $X_1, X_2, \dots, X_n$ are logical propositions and $t \in \{0, 1\}^n$, write $\vec{X}_t$ to be the conjunction of the logical propositions, with X_i negated iff $t_i = 1$.

Definition 2. If $X_1, X_2, \dots, X_n$ are logical propositions and $t \in \{0, 1\}^n$, write $\text{ALL}(X \leftarrow t)$ to be the conjunction of those propositions, but with X_i negated if $t_i = 1$. That is,

$$\text{ALL}(X \leftarrow t) = \bigwedge_{i \in [1, n]} Y_i, \text{ where } Y_i = \begin{cases} X_i, & \text{if } t_i = 0, \\ \neg X_i & \text{if } t_i = 1. \end{cases}$$

Useful logical fact 2 If $X_1, X_2, \dots, X_n$ are logical propositions and $t \in \{0, 1\}^n$, the following generalisation of logical fact 1 is true:

$$\bigwedge_{t \in \{0, 1\}^n} Y_t \implies \bigvee_{t \in \{0, 1\}^n} (\text{ALL}(X \leftarrow t) \wedge Y_t)$$

Proof: one of the assignments τ of negations to $\vec{X}$ makes the whole conjunction true (it's just the one that has a 1 in each index i for which X_i is false). We're given Y_t is true for all t , so Y_τ is true, hence $\text{ALL}(X \leftarrow \tau) \wedge Y_\tau$ is true, so the whole disjunction on the RHS is true.

4.2.1. Algorithm for relaxing threshold assertions

This leads us to the following simple algorithm for dealing with uncertainty about thresholds:

4.2.2. Algorithm for threshold uncertainty

1. Initialise the assertion set with $AT(P)$ assertions for all $P \in AT$ and, similarly, $BT(P)$ for all $P \in BT$.
2. For all $2^{|ET|}$ possible assignments of AT or BT to the parties in ET , calculate the DH assertions from Proposition 1 and add the $DH(,)$ assertions to the assertion set. (Do not include AT or BT assertions for parties in ET .)
3. For each party P , its maximum and minimum seat counts are the maximum and minimum values encountered at any point in Step 2.
4. Return the seat count bounds and the assertion set.

This algorithm is correct, but we will improve upon it in two ways. First, we need to deal with the case that some DH assertions are infeasible—this is in the next section.

Second, we can show that it is not necessary to check all $2^{|ET|}$ possible assignments, but only the cases in which

1. all $P \in ET$ are above the threshold,
2. all $P \in ET$ are below the threshold,
3. for each $P \in ET$, P is above the threshold and all other parties in ET are below.

This is proven in LEAN in `RelaxThresholds.lean`. [There's a proof in the LEAN repo but it's difficult for humans to read.](#) Unclear whether we care or we should just do the $2^{|ET|}$ thing as John has been doing for a while. vT

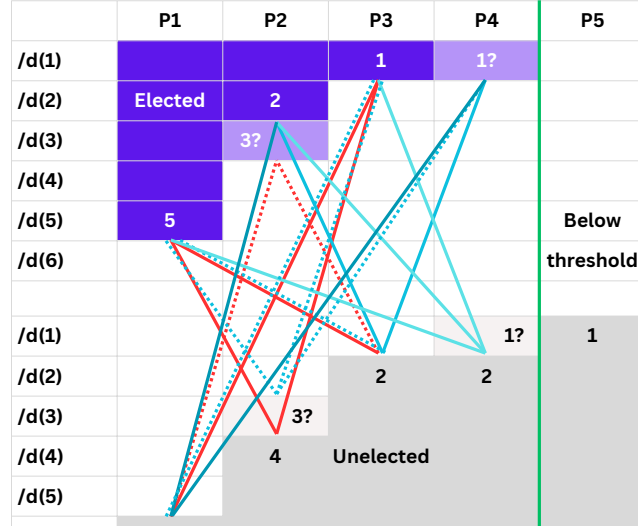


Figure 4: Illustration of the assertions to use when it is infeasible to audit whether a particular party (in this case P4) is above or below the threshold. The example shows what happens when either (P4 is above the threshold and P2 gets only 2 seats) or (P4 is below the threshold and P2 gets 3 seats). Blue lines are relevant to the first disjunct, red lines to the second. Redundant assertions are dotted.

Appendix B.1 contains the exact assertion sets and some simplifications. Figure 4 shows an example. This is simpler than it looks because many assertions are either repeated, or logically implied by others and hence redundant.

4.3. Relaxing in the case that a DH assertion is infeasible

When there is an infeasible DH assertion $DH_{A,B}(w,l)$, we can omit it and generate the assertions that allow the seat to be assigned to either side of the infeasible DH comparison. The assertions are given in Proposition 4 in Appendix B.1.1. This means adding an assertion that the second-lowest winner of Party A still defeated B's highest loser, and that the lowest winner of Party A defeated B's second-highest loser. Figure 5 shows an example—the added assertions are shown in blue.

(Note there is no guarantee that the lowest-margin DH assertion impacts either the overall highest loser or lowest winner.)

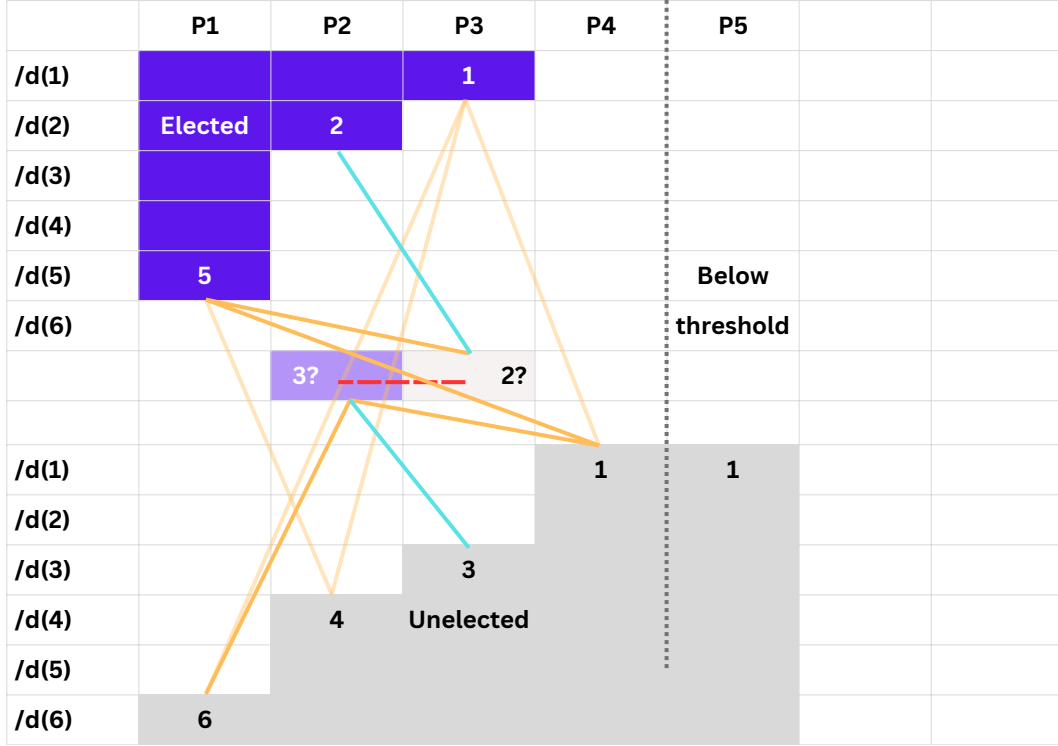


Figure 5: Illustration of the assertions to use when it is infeasible to audit a DH assertion. The example shows what happens when the DH assertion that P2’s 3rd candidate defeats P3’s 2nd candidate is relaxed—this assertion (which is removed) is shown as a dashed red line. Blue lines show the relaxed DH assertions that replace it. The other assertions stay the same—these are shown in orange. We leave P2’s 3rd candidate and P3’s second candidate in an “unsure” category. Note there does not have to be an assertion between P1’s 5th candidate and P3’s 3rd candidate, because this holds transitively. Similarly P2’s 2nd vs P3’s 3rd.

This process can be iterated if there are multiple infeasible DH assertions.

4.3.1. Relaxing multiple DH comparisons

DH winner-loser pairs can be relaxed iteratively, by maintaining the following counts for each party:

- $\mathbf{DH}_W$ clear DH winners,
- $\mathbf{DH}_U$ uncertain DH outcomes,
- $\mathbf{DH}_L$ clear DH losers.

With no relaxations, $DH_W(p) = s(p)$ and $DH_L(p) = \|p\| - s(p)$ for all $p \in \mathcal{P}$. In the example in Figure 5, $DH_W(P2) = 2$, $DH_U(P2) = 1$, $DH_W(P3) = 1$ and $DH_U(P3) = 1$. Party A 's first clear loser is $DH_W(A) + DH_U(A) + 1$. Its first uncertain candidate, if it exists, is $DH_W(A) + 1$. Its last uncertain candidate, if it exists, is $DH_W(A) + DH_U(A)$.

This suggests the following algorithm (ignoring thresholds).

4.3.2. DH-relaxing algorithm

1. Generate assertions for the apparent outcome, as in Proposition 1.
2. Calculate sample sizes for the DH assertions. If all are acceptable, RETURN those assertions and the exact seat allocation.
3. Initialize $DH_W(p) = s(p)$ and $DH_U(p) = 0$, for all $p \in \mathcal{P}$.
4. For all unacceptable DH assertions $DH_{A,B}(w, l)$ in increasing order of margin,³
 - a) Remove $DH_{A,B}(w, l)$ from the assertion set,
 - b) If $w > 1$, add $DH_{A,B}(w - 1, l)$ to the assertion set,
 - c) If $DH_W(A) \geq w$, set $DH_W(A) = w - 1$ and increment $DH_U(A)$. This moves A 's w -th quotient from the winning set into the uncertain set. (If w was already uncertain, do nothing.)
 - d) If $l < |B|$ (i.e. if B has any other losers), add $DH_{A,B}(w, l + 1)$ to the assertion set,
 - e) If $DH_W(B) + DH_U(B) < l$, set $DH_U(B) = l - DH_W(B)$. This moves B 's l -th quotient from the losing set into the uncertain set. (If l was already uncertain, do nothing.)

If all are acceptable, RETURN those assertions and the corresponding bounds ($DH_W(P)$ to $DH_W(P) + DH_U(P)$).

5. Go to Step 4.

VT

In steps 4c and 4e, I can't see how the comparison would ever be *strictly* true, and I'm not sure it's correct if it is - I think this would only happen if we're relaxing multiple DH assertions for the same party, with different winners or losers. In this case, I hope that the bounds-update is correct when we sort increasing by margin, but it needs a proofread. Devlin says the strict comparison never happens and so the sort-by-margin preprocess is unnecessary.

³VT: This might not be exactly right in the case where we're relaxing multiple DH assertions for the same party - the idea is right, but the conditions on exactly when to increment or decrement the bounds probably deserve a proofread. I'm hoping that processing in increasing order of margin prevents anything going wrong.

4.4. Relaxing threshold and DH assertions

The aim is to derive bounds for the number of seats allocated to each party. We choose some extreme scenarios that give worst- and best-case scenarios for parties in certain classes. For all these scenarios, a DH relaxation step can also be done. This provides upper and lower bounds on the number of allocated seats for each party.

The following algorithm combines a shortcut version of 4.2.2 with the DH-relaxing algorithm in 4.3.2. Again, it is correct to use all $2^{|ET|}$ possibilities for above/below threshold values in ET rather than the shortcut shown here.

4.4.1. Complete algorithm

1. Include $AT(P)$ assertions for all $P \in AT$ and, similarly, $BT(P)$ for all $P \in BT$.
2. Assume all $P \in ET$ are *above* the threshold (but do not add AT assertions for them).
 - a) Run the algorithm from subsection 4.3.1 to derive relaxed DH assertions and the corresponding relaxed-DH values—call them $DH_W^{\text{low}}(A)$, $DH_U^{\text{low}}(A)$, $DH_L^{\text{low}}(A)$.
Call the corresponding assertion set the *Low Assertion Set*
3. Assume all $P \in ET$ are *below* the threshold (but do not add BT assertions for them).
 - a) Run the algorithm from subsection 4.3.1 to derive relaxed DH assertions and the corresponding relaxed DH values $DH_W^{\text{high}}(A)$, $DH_U^{\text{high}}(A)$, $DH_L^{\text{high}}(A)$.
Call the corresponding assertion set the *High Assertion Set*
4. For each $P \in ET$, assume that P is above the threshold and all other parties in ET are below it (but do not add any AT or BT assertions).
 - a) Run the algorithm from subsection 4.3.1 to derive relaxed DH assertions and the corresponding relaxed DH values $DH_W^{\text{P-solo}}(A)$, $DH_U^{\text{P-solo}}(A)$, $DH_L^{\text{P-solo}}(A)$.
Call the corresponding assertion set the *P-solo Assertion Set*
5. If for any $P \in BT$, it is harder to audit $BT(P)$ than to prove that P is a DH loser to at least s other AT quotients, remove $BT(P)$ and add the necessary DH assertions instead.

VT

From Devin: Step 5 of the complete algorithm ("DH loser to at least S others"): the Lean verifier only counts quotients of AT rivals towards an upper bound of 0 (ET rivals may be ineligible), and it needs the per-pair form $DH_{Q,P}(s_Q, 1)$ for every winner Q in each world—not an arbitrary S quotients. Worth stating precisely. VT: now fixed.

4.4.2. Summary/implications

If $P \in AT$, then $DH_W^{\text{low}}(P)$ are P 's clear winners even under the most pessimistic scenario (for P) of threshold-uncertain parties meeting the threshold, and even after all the DH relaxations. The most optimistic scenario (for P) is that all uncertain parties fail the threshold, then P gets all its DH-uncertain seats in that case.

This gives us the following bounds:

$$\forall P \in AT, DH_W^{\text{low}}(P) \leq s(P) \leq DH_W^{\text{high}}(P) + DH_U^{\text{high}}(P).$$

If $P \in ET$, then it could be below the threshold and get 0 seats. Its maximum occurs when it is the sole above-threshold party out of all the uncertain parties, and it gets all its DH-uncertain seats in that case. Hence:

$$\forall P \in ET, 0 \leq s(P) \leq DH_W^{\text{P-solo}}(P) + DH_U^{\text{P-solo}}(P).$$

Obviously if $P \in BT$ then $s(P) = 0$.

This gives a complete characterisation of the bounds on every party's true seat allocation.

4.4.3. Summary of Assertions

TODO: Write out explicitly what the disjunction of these two sets of assertions is. Possibly we can simplify a bit, as in Section 4.2.

From subsubsection 4.4.1, we have

- $AT(P)$ assertions for all $P \in AT$ and, similarly, $BT(P)$ for all $P \in BT$,
- the low assertion set,
- the high assertion set,
- the P-solo assertion set for each $P \in ET$.

5. Other random not-finished ideas

VT

This part isn't finished. Possibly this should eventually be in the intro. Don't bother reading too carefully below here.

We are given:

- The election context (validity rules, social choice function, etc - to be defined more precisely)
- All the setup from subsection 2.3, including the tallies and parties

And we have some mutually-dependent variables to work with:

- the risk limit α ,

- the error rate e ,⁴
- a set of assertions $\mathcal{A}$,
- the minimum (apparent) margin $\mu_{\min}$ over $\mathcal{A}$,
- the estimated sample size σ .

We could consider these variables to be dependent in various directions. For example, in a traditional RLA, we are given α and asked to generate some assertions that imply the outcome, then estimate the sample size.

Problem 1 (Standard RLA).

Given: α and e ,

Generate: *Assertions $\mathcal{A}$ that imply the announced election outcome, with a sample-size estimate σ .*

VT

TODO: This isn't quite right because I want to distinguish between the cases where we're talking about static setup (e.g. maximin in the style of RAIRE) vs the case where we're talking about dynamic estimates (e.g. AWAIRE or risk-measuring audits).

Problem 2 (Risk-measuring audit). ***Needs the actual audit data, not just an estimate in advance.*

VT

Problem 3 (Flipped RLA). *TODO: Obvious flipped version in which you're given a maximum estimated sample size and are asked to generate assertions, and to report the minimum value of α with at most that estimated sample size.*

Since auditing all the allocations of a Belgian election might be infeasible, we could instead frame the question in terms of trying to generate an assertion set that is feasible to audit but may not necessarily imply the exact election outcome.

Problem 4 (Relaxed RLA).

Given: α , an (estimated) error rate e , and a maximum sample size σ ,

Generate: *Relaxed assertions $\mathcal{A}$ that imply "the closest approximation we can test"⁵ to the announced election outcome, s.t. risk limit α can be met in σ samples given error rate e .*

Problem 5 (Relaxed RLA - decision version). **TODO**

=====
What follows is from an old version of these notes.
=====

⁴Possibly we will need to care about more specific details of the errors rather than just the standard rate.

⁵To be defined

6. Some simple observations

We now define some important special winners.

Definition 3. *The last winner w_{Last} is the winner whose quotient is the least of all winners.*

Proposition 2. *If $t(w_{Last}) < \theta$ then, for any loser l whose party is below the threshold, it is always more efficient to prove $BT(P_l)$ than $DH-Loser(l)$.*

Proof needs some care.

Note that the opposite is not necessarily true: if $t(w_{Last}) \geq \theta$ then it might be more efficient to prove $BT(P_l)$ or $DH-Loser(l)$.

It is obvious that if a party is above the threshold then the only option is to prove $DH-Loser(l)$ for their first loser. This therefore gives a complete recipe for the most efficient audit in the case that $t(w_{Last}) < \theta$.

Conjecture 1. *If $t(w_{Last}) < \theta$ then the most efficient auditing strategy is:*

- for all parties P with at least one winner, prove $AT(P)$,
- for all parties P for which $BT(P)$ is true, prove $BT(P)$,
- for all parties P for which $BT(P)$ is false that have at least one loser, prove $DH-Loser(l)$ for P 's first loser l .

If $t(w_{Last}) \geq \theta$, then we are in a more difficult situation.

Conjecture 2. *if $t(w_{Last}) \geq \theta$, then an “efficient” auditing strategy could be:*

- for all parties P with at least one winner, prove $AT(P)$,
- for all parties P with at least one loser:
 1. Compute the minimum sample size for $BT(P)$
 2. Compute the minimum sample size for $DH-Loser(s((\cdot) + 1P))$

prove whichever of these assertions is cheaper

7. Margin of a coalition

VT: Possibly this is superseded by the new stuff in the next section.

Here are some quick notes on one heuristic strategy to evaluate the vote margin of a coalition C .

Let m_s be the seat margin of that coalition in terms of seats, that is, if the reported result show that this coalition got x seats, then $m_s = x - 75$.

We iterate over all possible ways in which C can lose m_s seats. That is, we consider every possible way to define $e(P)$ for every $P \in C$ such that $\sum_{P \in C} e(P) = m_s$.

When $e(P)$ is defined, we iterate over every party in $P \in C$ such that $e(P) > 0$ and count the minimum number of votes that each P should lose to a party not in C in order to lose a seat. We keep the minimal value and call this this margin of the first seat, and decrease by 1 the value of $e(P)$ for this "weakest" P . (That P may either be the party with the lowest winning quotient within the coalition, or the one closest to θ .)

As long as $e(P)$ is not null everywhere, we iterate: after having removed the votes from the 1st weakest party, we start removing votes from the 2nd weakest party and see how many extra votes need to be removed in order to make the coalition lose a second seat. We continue in this way until we removed m_s seats from C , and define the vote margin as the total number of votes we had to remove.

The vote margin of the coalition will be the minimum vote margin taken over every possible way of defining the e function for that coalition.

Question: does this strategy, which proceeds by grinding every weakest seat one after the other, an optimal strategy to obtain the minimum number of votes that need to be changed in order to break a winning coalition in a way that is consistent with a specific choice of e ?

8. Safe subsets of apparently-winning coalitions

We want to examine when it is safe to audit a relaxation of an apparent outcome, rather than the exact outcome itself.

Define $\text{SAFE-SUBSET}(s_1, s_2)$ to be true if the winning coalitions under s_1 are identical to those under s_2 . A coalition wins if it gets $\lfloor S/2 \rfloor + 1$ seats. A correct (though inefficient) implementation is given in 1.

Algorithm 1 SAFE-SUBSET

For testing that two different seat allocations have the same winning coalitions

```

1: Input: set of parties  $\mathcal{P}$ , number of seats to be allocated  $S$ .
2: Input: seat allocations  $s_1$  and  $s_2$ .
Require:  $s_2$  is a sub-allocation of  $s_1$ , that is,  $\forall p \in \mathcal{P}, |s_2(p)| \leq |s_1(p)|$ .
3: for all coalitions  $C \in 2^{\mathcal{P}}$  do
4:    $t_1 := \sum_{p \in C} s_1(p)$  ▷ The coalition's tally under  $s_1$ 
5:    $t_2 := \sum_{p \in C} s_2(p)$  ▷ The coalition's tally under  $s_2$ 
6:   if  $t_1 \geq \lfloor S/2 \rfloor + 1 \wedge t_2 \leq \lfloor S/2 \rfloor$  then
7:     return false ▷ C wins under  $s_1$  but not  $s_2$ 
8:   end if
9:   if  $t_1 \geq \lceil S/2 \rceil \wedge t_2 \leq \lceil S/2 \rceil - 1$  then
10:    return false ▷  $\mathcal{P} \setminus C$  wins under  $s_2$  but not  $s_1$ 
11:  end if
12: end for
13: return true

```

9. Auditing a relaxation of an apparently-winning coalition

Suppose that instead of verifying the exact apparent allocation s_{AW} (the ‘apparent winners’) of seats to parties, we instead wish to verify a sub-allocation s_{CW} (the ‘clear winners’).

9.1. Defining the relaxation

When we calculate margins, we may have a number of assorter margins that seem too small for auditing efficiently. We’re going to try to ignore some of them, but we need to check that this preserves winning coalitions. We still need to prove that the ‘clear winners’ CW really won. This may need to be done iteratively, by identifying the closest margin, trying to ignore it, seeing whether the resulting relaxation is valid and auditable, and then either reverting, accepting, or trying to remove the next-closest assertion. For this section, define:

AW (‘apparent winners’) - parties that have at least one winner, according to the official tally;

AL (‘apparent losers’) - parties that have at least one loser, according to the official tally;

$\mathcal{U}_{AT}$ (‘uncertain above threshold’) - parties that are apparently above the threshold, but we are going to avoid testing that they are (because of a small margin for the AT assorter);

$\mathcal{U}_{BT}$ (‘uncertain below threshold’) - parties that are apparently below the threshold, but we are going to avoid testing that they are (because of a small margin for the BT assorter);

These sets are not necessarily distinct.

Also assume we are given *Uncertain D’Hondt* $\mathcal{U}_{DH}$, a set of (individual candidate) apparent winners (P, i) (candidate number i in party P) for whom at least one DH assorter $DH_{P, P_L}(\cdot)$ is very small. We will also try to avoid testing whether P got i seats, instead trying to see if it is enough to prove that P got $i - 1$.

s_{CW} is derived from the apparent allocation s_{AW} by

- removing any allocation to parties in $\mathcal{U}_{AT}$, and
- reducing by 1 the allocations to $\mathcal{U}_{DH}$.

The main idea is to test that *at least* the allocations in s_{CW} are true, while remaining ambivalent about whether the parties in $\mathcal{U}_{AT}$ and $\mathcal{U}_{BT}$ are actually above or below the threshold.

9.2. Assumptions and sanity checks

The following should be checked before the audit starts—they are assumptions about the setup.

1. The relaxation does not change the winning coalitions, that is

$$\text{SAFE-SUBSET}(s_{AW}, s_{CW}).$$

2. s_{CW} is a sub-allocation of s_{AW} , that is

$$\forall p \in \mathcal{P}, s_{CW}(p) \leq s_{AW}(p).$$

3. s_{AW} matches the expected number of available seats, that is

$$\sum_{p \in \mathcal{P}} s_{AW}(p) = S.$$

4. There is at least one winning coalition under CW, that is

$$\sum_{p \in \mathcal{P}} s_{CW}(p) \geq \lfloor S/2 \rfloor + 1.$$

5. The second candidates in parties in $\mathcal{U}_{AT}$ and $\mathcal{U}_{BT}$ are apparent D'Hondt losers wrt the apparent winners, that is $\forall p_W \in AW, \forall p_L \in \mathcal{U}_{AT} \cup \mathcal{U}_{BT}$,

$$DH_{p_W, p_L}(s(p_W), 2)$$

(Note this is an assumption about the apparent tallies, not an assertion to be verified, though similar assertions will indeed be verified—see below.)

6. The ‘clear winners’, plus the first candidates of the threshold-uncertain parties, are no more than the allowed number of seats, that is

$$|\mathcal{U}_{AT}| + |\mathcal{U}_{BT}| + \sum_{p \in \mathcal{P}} s_{CW}(p) \leq S.$$

This is a necessary sanity check on ‘clear winners’, because the last D'Hondt winner (from a large party) usually has a quotient much smaller than θ , so in order for the ‘clear winners’ to be what they sound like, we need to ensure that ambivalence about who is over the threshold does not give us more than the allowed number of winners—if it did, the ‘clear winners’ would drop out of the D'Hondt winning set.

9.2.1. What assertions to prove

Now we want to prove that (at least) all the seats allocated by s_{CW} are correct. Let CW be the set of parties with at least one seat allocated by s_{CW} , and for a party $p \in CW$, let the *lowest clear winner* $lcw(p)$ be the number allocated to p by s_{CW} .

Then it suffices to test the following assertions:

1. $\forall p_{CW} \in \mathcal{CW}$,

$$AT(p_{CW})$$

2. **2.1** $\forall p_{CW} \in \mathcal{CW}, \forall p_L \in \mathcal{U}_{\mathcal{BT}} \cup \mathcal{U}_{\mathcal{AT}}$,

$$DH_{p_{CW}, p_L}(lcw(p_{CW}), 2))$$

This used to say $DH_{p_{CW}, p_L}(lcw(p_{CW}, 2))$ but I think the parentheses were in the wrong place. Ditto below. VT

2.2 $\forall p_{CW} \in \mathcal{CW}, \forall p_L \in \mathcal{AL} \setminus \{\mathcal{U}_{\mathcal{BT}} \cup \mathcal{U}_{\mathcal{AT}}\}$,

$$BT(p_L) \vee DH_{p_{CW}, p_L}(lcw(p_{CW}), s(p_L) + 1)$$

where $s(p_L) + 1$ is defined by the (official) *apparent outcome* s_{AW} .

TODO prove that this set of assertions is sufficient to imply that the allocations in s_{CW} are a sub-allocation of the true allocation, given the assumptions and sanity checks in the previous section. Note it's important that $\mathcal{U}_{\mathcal{BT}} \cup \mathcal{U}_{\mathcal{AT}}$, even if they all win one seat, don't cause CW to exceed S .

10. Coalitional Audits

MB (Note: Not sure if this is covered above, or if this is a new/slightly different approach).

Let $\mathcal{C}$ denote the set of all valid coalitions that could form on the basis of the reported results, where each $C \in \mathcal{C}$ is itself a set of parties. A set of parties C is a valid coalition on the basis of the reported results iff the total number of seats awarded to parties in C is at least 76.

Let R denote the set of circumscriptions, indexed by j . Let $\mathcal{P}$ denote the set of parties, indexed by i . Let $N_{i,j}$ denote the number of seats awarded to party i in circumscription j .

A coalitional audit verifies, for each $C \in \mathcal{C}$, that C won at least 76 seats across the set of circumscriptions.

For each $C \in \mathcal{C}$ we must define a $n_{i,j} \in \{0 \dots N_{i,j}\}$ for each party i in coalition C and circumscription j in which party i is present, such that:

$$\sum_{j \in R} \sum_{i \in C} n_{i,j} \geq 76 \quad (3)$$

What we need to show:

- For each $j \in R$ and $i \in C$ for which $n_{i,j} > 0$, party i got more than 5% of the vote in circumscription j (i.e., they were eligible to receive seats).
- That the $(n_{i,j})^{th}$ winner in party i in circumscription j beats the highest loser in each party $k \neq i$ present in circumscription j . (I believe this can be done with DH assertions).

The problem becomes, for each $C \in \mathcal{C}$:

- Selecting a value for $n_{i,j}$ for each $i \in C$ and $j \in R$ such that the cost of the collection of assertions is minimised.

A. Estimating which disjunct requires fewer samples

This section discusses possible heuristics for choosing which of $BT(B)$ or $\bigwedge_{A \in \mathcal{W}} DH_{A,B}$ to audit.

(we could verify both, but that appears to be a waste of resources).

We discuss some heuristics here. Let us define $DH\text{-}Loser(B)$ for $B \in \mathcal{L}$ the assertion that $\bigwedge_{A \in \mathcal{W}} DH_{A,B}(.,.)$

A.1. Check margins

If θ is the electoral threshold, we can compute $m_1 = \theta - \mathcal{Q}(B, s(B) + 1)$ and m_2 to be the minimum over all $A \in \mathcal{W}$ of the number of votes that would need to be switched from list A to list B in order to make the first loser of B win over the last winner of A .

For parties $B \in \mathcal{L}$, we have a choice: do we test $BT(B)$ or all $DH_{A,B}(f)$ or every $A \in \mathcal{W}$, i.e., $DH\text{-}Loser(B)$. This choice comes down to which one we think are easiest to verify. We can estimate this by using the reported margins of each assorter. Let m_1 be the assorter margin of $BT(B)$:

$$m_1 = \theta \times V - \mathcal{T}(B)$$

Let m_2 be the assorter margin of the D'Hondt assertions:

$$m_2 = \min_{A \in \mathcal{W}} \mathcal{Q}(A, s(A)) - \mathcal{Q}(B, s(B) + 1)$$

Then, if $m_1 > m_2$, we test $BT(B)$ and we test $DH\text{-}Loser(B)$ otherwise. If $m_1 < 0$ B was reported below the threshold. However, m_2 cannot be below 0 (in that case, reported result does not match CVRs) and is 0 when there is a reported tie.

Question: Not sure whether this may not be optimal, especially when there are many parties, because $DH\text{-}Loser(B)$ is big conjunction of assertions and, even if each assertions is easier to verify than $BT(B)$, we need to check many of them. On the other hand, we check them all in parallel, so the expected number of assertions may not change, but just the standard deviation around that expectation?

True. The computational cost probably doesn't matter, because at worst it's $O(p^2)$ where p is the number of parties. However, the expected sample size of a big conjunction is indeed higher than the expected sample size of the worst thing, so (like RAIRE) this is an approximation, and I agree that in this case it might not be a very good one. That is, if we had a BT assertion with margin μ , and a collection of lots of DH assertions with margin $\mu + \epsilon$, we're saying at the moment that we should take the DH option, but actually it's probably better to take the BT option. Actually now I see someone has sketched this below.

AE unless an AWAIRE style method is used, which takes into account that only one needs to be verified AE Is there something missing here? I'll try and restate it below (if I understood correctly)

AE Computationally, there is a higher cost of testing $DHondtLoser$ as there are more assertions to test. However, you're posing an interesting question with regards to statistical efficiency. The number of assertions should not depend

A.2. Compare expectations based on ALPHA martingales

We can write assertions for $BT(B)$ and $DH\text{-}Loser(B)$ and estimate how many ballots we would need to sample for verifying each of these two assertions. Then we decide to pick the one that offers the smallest expectation on the number of ballots to sample.

Question: Is this any better than relying on the margins as before? Yes, I think if we could do this carefully then it's probably better. Could be done either on-the-fly AWAIRE style, or by simulation in advance. However, I'd be surprised if it makes a huge difference because the bad cases will probably be dominated by one bad assertion anyway.

Furthermore, are such local estimations good estimates or what will happen when they are composed? More precisely, if C is (a set of) assertions, is it the case that if Assertion A is cheaper to verify than Assertion B , then verifying A and C is cheaper than verifying B and C ? I would conjecture that this conjecture is correct in the present case because all the assertions are related to a single race, so everything is checked in parallel and, in the worst case, the cost of verifying C will dominate everything?

Sorry don't understand - why is it the disjoint sum rather than set union?

A.3. Compare expectations based on RAIRE (or) AWAIRE)

I understand that AWAIRE would remove the need to decide in advance which set of assertion to check and would rather choose adaptively based on the ballots that are sampled. But this is only useful when CVRs are not available, which is not the case in Belgium.

Question: Does RAIRE bring any benefit over ALPHA in the D'Hondt case here? I have no idea of how hard it would be to adapt it to the D'Hondt setting.

I'd reserve the term RAIRE for IRV, and say "choosing the assertions that maximin margin in advance" but agree with AEK - either should work. maximin is simpler; AWAIRE is better if you have enough errors that the optimal assertions are actually different from what the CVRs told you would be optimal. We could also do a more careful job of estimating the sample size of those conjunctions in advance—agree that this would be better, and that AWAIRE might do it automatically (but it might still be better to do it in advance). OTOH if there's only one small-margin assertion then that will dominate, so the extra complexity might not be worth it.

B. Explicit relaxed assertion sets

This appendix gives explicit descriptions of the assertions required for relaxed sets. Very unclear that this is useful - the algorithmic descriptions are easier to understand and more likely to be correct.

Assumption: R is either the lowest party above the threshold, or the highest party below the threshold, according to apparent tallies. Corollary: Regardless of whether we

VT

AE Just re-
alised some-
thing... For
 $DHondtLoser(I$
we estimate
how many
ballots for
 B we need
for each
 $DHondtA, B$
and how
many *mu-
tually
exclusive*
ballots for
each A we
need. Then
we can esti-
mate overall
sampling
effort for
 $DHondtLoser(I$
by taking
the maxi-
mum ballots
needed for
 B and sum
the number
of ballots
needed for
every A .
This should
be more
robust than
just taking
maximum
margin
of each
 $DHondt$
individually

assume $AT(R)$ or $BT(R)$, all (other) parties with a winner have a higher apparent tally than R .

Let σ be the seat allocation if $AT(R)$ is true, and let $x : \mathcal{P} \rightarrow \mathbb{N}$ be the function that gives extra seats to parties if $AT(R)$ is false. That is, the apparent allocation applied when we believe $AT(R)$ is false would give $\sigma(p) + x(p)$ to each party p .

Some quick observations:

- $\sum_{p \in \mathcal{P}} x(p) = 0$ because the total number of seats allocated is constant.
- $\forall p \neq R, x(p) \geq 0$ because when R drops below the threshold, only R loses seats.

B.0.1. Case 1: $\sigma(R) = 0$

In this case, R doesn't get any seats even if it is above the threshold. TODO. I suspect this case isn't actually very interesting. Possibly we just need to identify it and not bother checking what we don't need to. VT

B.0.2. Case 2: $\sigma(R) \geq 1$

Lemma 1. $x(p) > 0$ only if $\sigma(p) > 0$. That is, only parties that already have seats gain seats when R drops below the threshold.

Proof. This relies on the assumption above that all parties above θ have a higher tally than R . If R gains seats when assumed to pass the threshold, then all other parties above the threshold must also have seats. (Note this lemma is also trivially true in Case 1.) $\square$

Let $\mathcal{W}$ be the set of parties with at least one winner according to σ and let $\tilde{\mathcal{W}}$ be the set of winners according to $\sigma + x$. By Lemma 1, this differs only in that R is removed. That is,

$$\tilde{\mathcal{W}} = \mathcal{W} \setminus \{R\}. \quad (4)$$

Let $\mathcal{L}_1$ be the set of parties with at least one loser according to σ ; let $\mathcal{L}_2$ be the set of parties with at least one loser according to $\sigma + x$. $\mathcal{L}_1$ may contain parties not in $\mathcal{L}_2$; the reverse is not true except that R is in $\mathcal{L}_2$ and may not be in $\mathcal{L}_1$. That is, we have

$$\mathcal{L}_2 \setminus \{R\} \subseteq \mathcal{L}_1. \quad (5)$$

Using Proposition 1, the assertions sufficient to prove the allocation assuming $AT(R)$ are:

$$AT(R) \quad (6)$$

$$AT(A) \quad \text{for all } A \in \tilde{\mathcal{W}} \quad (7)$$

$$BT(B) \vee \left(\bigwedge_{A \in \mathcal{W}} DH_{A,B}(\sigma(A), \sigma(B) + 1) \right) \quad \text{for all } B \in \mathcal{L}_1. \quad (8)$$

Call this “Assertion set 1.”

Again using Proposition 1, the assertions sufficient to prove the allocation assuming $BT(R)$ are:

$$\begin{array}{ll} AT(A) & \text{for all } A \in \tilde{\mathcal{W}} \\ BT(B) \bigvee \left(\bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A) + x(A), \sigma(B) + x(B) + 1) \right) & \text{for all } B \in \mathcal{L}_2 \end{array}$$

We are assuming for this case that $\sigma(R) \geq 1$, so R ’s first candidate is a D’Hondt winner. The right disjunct in the second clause when $B = R$ must therefore be false, Right? I hope that’s obviously correct rather than subtly wrong. so this set of assertions VT must be proven as:

$$BT(R) \tag{9}$$

$$AT(A) \quad \text{for all } A \in \tilde{\mathcal{W}} \tag{10}$$

$$BT(B) \bigvee \left(\bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A) + x(A), \sigma(B) + x(B) + 1) \right) \quad \text{for all } B \in \mathcal{L}_2 \setminus R \tag{11}$$

Call this “Assertion set 2.”

We can then use the useful logical fact to establish a sufficient set of assertions to prove the relaxed claim.

Proposition 3. *In a D’Hondt election in which it is infeasible to check whether Party R passed the threshold, let σ be the apparent seat allocation when R passes the threshold and $x(P)$ be the extra seats allocated to party $P \in \mathcal{P}$ if R fails the threshold. Then the following assertions imply that for all $P \in \mathcal{P}$, P received at least $\sigma(P)$ and at most $\sigma(P) + x(P)$ seats.*

$$AT(A) \quad \text{for all } A \in \tilde{\mathcal{W}} \tag{12}$$

$$BT(B) \bigvee \left(\bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A), \sigma(B) + 1) \right) \quad \text{for all } B \in \mathcal{L}_1. \tag{13}$$

$$BT(B) \bigvee \left(\bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A) + x(A), \sigma(B) + x(B) + 1) \right) \quad \text{for all } B \in \mathcal{L}_2 \setminus R \tag{14}$$

Proof. Apply useful logical fact 1 to assertion sets 1 and 2, with $X = AT(R)$ and $\neg X = BT(R)$. □

B.1. Simplifying the assertions from 4.2

Using the “useful logical fact” with $X = AT(R)$, Y as assertion set 1 and Z as assertion set 2, the disjunction of assertion sets 1 and 2 is implied by:

$$\begin{array}{ll}
 AT(A) & \text{for all } A \in \tilde{\mathcal{W}} \\
 BT(B) \vee \left(\bigwedge_{A \in \mathcal{W}} DH_{A,B}(\sigma(A), \sigma(B) + 1) \right) & \text{for all } B \in \mathcal{L}_1 \\
 BT(B) \vee \left(\bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A) + x(A), \sigma(B) + x(B) + 1) \right) & \text{for all } B \in \mathcal{L}_2 \setminus R
 \end{array}$$

Since $\mathcal{L}_2 \setminus \{R\} \subseteq \mathcal{L}_1$, (Equation 5), this is implied by

$$\begin{array}{ll}
 AT(A) & \text{for all } A \in \tilde{\mathcal{W}} \\
 BT(B) \vee \left(\bigwedge_{A \in \mathcal{W}} DH_{A,B}(\sigma(A), \sigma(B) + 1) \right) & \text{for all } B \in \mathcal{L}_1 \\
 BT(B) \vee \left(\bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A) + x(A), \sigma(B) + x(B) + 1) \right) & \text{for all } B \in \mathcal{L}_1
 \end{array}$$

(Remember that $DH_{A_i, B_j}(\cdot, \cdot)$ s vacuously true if B doesn't have j candidates.)

Rewriting $\mathcal{W}$ as $\tilde{\mathcal{W}} \oplus \{R\}$ (by Equation 4):

$$\begin{array}{ll}
 AT(A) & \text{for all } A \in \tilde{\mathcal{W}} \\
 BT(B) \vee \left(DH_{R,B}(\sigma(R), \sigma(B) + 1) \wedge \bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A), \sigma(B) + 1) \right) & \text{for all } B \in \mathcal{L}_1 \\
 BT(B) \vee \left(\bigwedge_{A \in \tilde{\mathcal{W}}} DH_{A,B}(\sigma(A) + x(A), \sigma(B) + x(B) + 1) \right) & \text{for all } B \in \mathcal{L}_1
 \end{array}$$

Useful logical fact 2: the distributive law If X, Y, Z are logical propositions,

$$(X \vee Y) \wedge (X \vee Z) \iff X \vee (Y \wedge Z)$$

We can use this to rewrite lines 2 & 3 (with $BT(B)$ as X), giving us:

$$BT(B) \vee \left(DH_{R,B}(\sigma(R), \sigma(B) + 1) \wedge \bigwedge_{A \in \tilde{\mathcal{W}}} (DH_{A,B}(\sigma(A), \sigma(B) + 1) \wedge DH_{A,B}(\sigma(A) + x(A), \sigma(B) + x(B) + 1)) \right)$$

B.1.1. Assertions for a relaxed DH assertion

Proposition 4. *In a Belgian federal election where the reported outcome assigns all the seats, we define $\mathcal{W}$ the set of parties that have at least one reported winner and $\mathcal{L}$ the set of parties that have at least one reported loser. Suppose the lowest-margin DH assertion is $DH_{R,Q}(s(R), s(Q) + 1)$. Then the reported outcome is correct except possibly that R should get one less seat and Q one more if we prove:*

$$AT(A) \quad \text{for all } A \in \mathcal{W} \cup \{Q\} \quad (15)$$

(Not shown in Figure 5.)

$$BT(B) \vee \left(\bigwedge_{A \in \mathcal{W}} DH_{A,B} \right) \quad \text{for all } B \in \mathcal{L} \setminus Q \quad (16)$$

(Orange in Figure 5; dotted if redundant.)

$$BT(Q) \vee \left(\bigwedge_{A \in \mathcal{W} \setminus R} DH_{A,Q} \right) \quad (17)$$

(Orange in Figure 5; dotted if redundant.)

$$BT(B) \vee DH_{Q,B}(s(Q) + 1, s(B) + 1) \quad \text{for all } B \in \mathcal{L} \setminus \{Q, R\} \quad (18)$$

(Green in Figure 5.)

$$BT(B) \vee DH_{A,R}(s(A), s(R)) \quad \text{for all } A \in \mathcal{W} \setminus \{Q, R\} \quad (19)$$

(Green in Figure 5.)

$$DH_{R,Q}(s(R), s(Q) + 2) \quad \text{(Blue in Figure 5.)} \quad (20)$$

$$DH_{R,Q}(s(R) - 1, s(Q) + 1) \quad \text{(Blue in Figure 5.)} \quad (21)$$

This might still be not quite right. TODO. Check vs Figure 5.

VT

Proof. TODO. Hopefully just the disjunction again. □

References

- [1] Michelle L. Blom, Jurlind Budurushi, Ronald L. Rivest, Philip B. Stark, Peter J. Stuckey, Vanessa Teague, and Damjan Vukcevic. Assertion-based approaches to auditing complex elections, with application to party-list proportional elections. In *Electronic Voting - 6th International Joint Conference, E-Vote-ID 2021*, volume 12900 of *Lecture Notes in Computer Science*, pages 47–62. Springer, 2021.

KAMER VAN VOLKSVERTEGENWOORDIGERS

KIESKRING ANTWERPEN

26 MEI 2019

VERKIEZING VAN 24 VOLKSVERTEGENWOORDIGERS

3	8	9	10	12	15	16	19	20	21	22	23
Open Vld	N-VA	VLAAMS BELANG	CD&V	PVDA	GROEN	sp.a	Volt	DierAnimal	PV&S	D-SA	Piratenpartij
1. LUYCKX Christine	1. JAMBOIR Jan	1. VAN OPIEKEN Joke	1. VERHOEVEN Sander	1. MERTENS Tom	1. CAUW Tom	1. HENDRICKX Romy	1. VERBILLEN Joke	1. DINGENS Fanny	1. BEN MYHANS Saskia	1. IL JAFFRY Randy	1. DE VOSST Joan
2. VERHAERT Marjolaine	2. VAN RIEL Vera	2. SAMIN Ellen	2. LARIN Nathalie	2. GAGMS Griet	2. BUST Kris	2. BERTILS Rita	2. VERBIST Vera	2. BRILERS Tine	2. TONRIS Saskia	2. OPVOLGERS	2. WILLEMS Joke
3. THYS Liesbeth	3. WOLLAERTS Bert	3. VAN LOMMEL Nathalie	3. VAN DEN BERGH Jef	3. TOUM Muel	3. VANDEN BERGHE Romy	3. EL MOUARI Rita	3. KONGE Anne	3. MOONS Anne	3. ABDOU Marceline	3. AMOUKRI Nawal	3. VAN PETERICHEM Kris
4. SLAVER Romy	4. DE WIT Sofie	4. VERHEY Romy	4. DIERICKX Rita	4. DIERICKX Rita	4. DIERICKX Rita	4. VAN TINGELON Rita	4. VAN STRAETEN Saskia	4. VERBILLEN Joke	4. GERRAET Rita	4. GERRAET Rita	4. COLPARENT Joke
5. D'AMORSEAU Alexandra	5. FRIEDLICH Wendy	5. FRANCIS Catherine	5. SOARES Wim	5. SOARES Wim	5. HERTOGHS Jan	5. VANDEN BERGHE Romy	5. DE WILCKX Saskia	5. AMON Saskia	5. D'AMORSEAU Alexandra	5. GERRAET Rita	5. LIEBENS Joke
6. LARIN Nathalie	6. VAN CAMP Vera	6. CORNELISSEN Griet	6. BERTMANS Ellen	6. MERTENS Tom	6. FRIEDLICH Wendy	6. HERTOGHS Jan	6. BERTMANS Ellen	6. BERTMANS Ellen	6. BERTMANS Ellen	6. BERTMANS Ellen	6. BERTMANS Ellen
7. VAN HOUTENDIJK Elly	7. MERTENS Tom	7. CLOPPERS Rita	7. VERBILLEN Joke	7. CLOPPERS Rita	7. CLOPPERS Rita	7. CLOPPERS Rita	7. CLOPPERS Rita	7. CLOPPERS Rita	7. CLOPPERS Rita	7. CLOPPERS Rita	7. CLOPPERS Rita
8. MARINCKX Rita	8. MERTENS Tom	8. MERTENS Tom	8. VAN DEN BERGH Jef	8. JOOS Wouter	8. JOOS Wouter	8. JOOS Wouter	8. JOOS Wouter	8. JOOS Wouter	8. JOOS Wouter	8. JOOS Wouter	8. JOOS Wouter
9. FRIEDLICH Wendy	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie	9. DE WIT Sofie
10. VAN OPIEKEN Joke	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom	10. MERTENS Tom
11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita	11. CLOPPERS Rita
12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita	12. MARINCKX Rita
13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine	13. LUYCKX Christine
14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom	14. VAN DER SCHOT Tom
15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom	15. VAN DER SCHOT Tom
16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom	16. VAN DER SCHOT Tom
17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom	17. VAN DER SCHOT Tom
18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom	18. VAN DER SCHOT Tom
19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom	19. VAN DER SCHOT Tom
20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom	20. VAN DER SCHOT Tom
21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom	21. VAN DER SCHOT Tom
22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom	22. VAN DER SCHOT Tom
23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom	23. VAN DER SCHOT Tom

Figure 1: The ballot for the Antwerpen district in the 2019 federal elections.

# of candidates	L_1	L_2	L_3	L_4
Party total:	180	96	150	18
$d = 2$	90	48	75	9
$d = 3$	60	32		6
$d = 4$	24			

Merged quotients: 180, 150, 96, 90, 75, 60, 48, 32, 24.

Figure 2: Illustration of the D'Hondt tallying process with electoral threshold. If there are 6 seats, L_1 gets 3, L_2 gets 1 and L_3 gets 2 seats. L_4 does not reach the electoral threshold of $(180 + 96 + 150 + 18) \times 0.05 = 22.2$, and its quotients are therefore not listed.

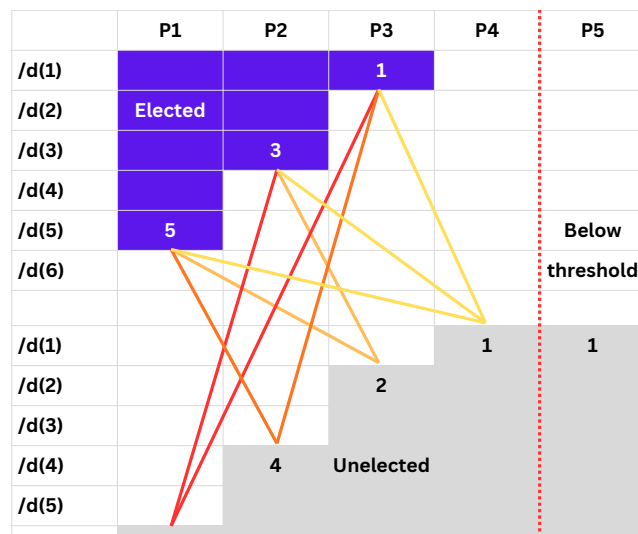


Figure 3: Illustration of D'Hondt and threshold assertions. In some elections, parties that are below the threshold may be D'Hondt winners.